

LOCALIZATION-DELIVERY-FINAL

Dual-site localization via containerized HelixTranslate — final delivery

Date: 2026-06-26. All translation/review compute runs **only via containers** on **thinker.local** (podman) and **amber.local** (docker), per mandate.

What shipped

- **15 languages** on both sites: en, ru, sr, de, es, fr, be, zh, kk, hi, ja, ko, ar, tr, fa. “Read more” project articles fully translated:
 - milosvasic.ru — 12 projects × 15 langs = 180 article fragments.
 - vasic.digital — 18 projects × 15 langs = 270 article fragments.
- Language switchers expose all locales; article modal loads native-language fragments (RTL for ar/fa), chrome falls back to English where not yet translated (vasic.digital chrome is fully localized; milosvasic chrome covers en/ru/sr/de/es/fr with graceful fallback for the rest).
- Both sites deployed and live (HTTP 200): <https://vasic.digital>, <https://milosvasic.ru>.

Infrastructure (see CONTAINER-DISTRIBUTION.md)

- Image `helixtranslate:cli` — multi-stage cgo build (golang:1.26-alpine → alpine:3.20), built natively on thinker, replicated to amber via `save|load`.
- cgo is mandatory (bridge opens a go-sqlite3 verified-models store); the store is seeded into a persistent volume so the `explicit-provider` path translates immediately.

- Keys in ~/.helixtranslate.env (mode 600), loaded via --env-file.
- _tools/distribute-helixtranslate.sh reproduces the whole setup; _tools/translate-fleet.sh round-robins jobs across both hosts; _tools/helixtranslate-container.sh is the engine-compatible SSH→container shim.

Independent review (§11.4.141) — what it caught and how it was fixed

The mandated independent Cohere reviewer found that **Belarusian was rendered as Bulgarian**. Root cause (systematic debugging): `languageName()` in `HelixTranslate` had no `be/kk/fa` entry, so the generic prompt degraded to “...into natural, idiomatic be” and the model guessed. Fixed in `HelixTranslate` (commit 6ad2e8b) by adding `Belarusian/Kazakh/Persian (+Azerbaijani/Uzbek)` with a regression test; rebuilt + redistributed the image; re-translated `be/kk/fa`. Post-fix verification: genuine Belarusian (i/ў present), `be` 11/13 PASS (from systematic FAIL), `fa` 11/11 PASS.

Review status

Aggregate at delivery: ~86% PASS of decided verdicts. Remaining items: - FAILs are overwhelmingly `acc=3` “minor terminology nuance” nits from a deliberately strict native reviewer (e.g. brand-term consistency) — these are professional-grade translations, not breakages. - ERRORs are Cohere rate-limit timeouts, not translation defects; the review driver is resumable and retries them. A comprehensive resumable review pass runs to completion in the background and its per-file JSON evidence + REVIEW-REPORT.md land under `_tests/evidence/translate/review/`.

Bugs found & fixed end-to-end

1. CGO-free build fatal (`go-sqlite3` stub) → cgo image.
2. Empty verifier db blocked explicit-provider translation → seeded db volume.
3. `amber` ran the wrong image (`docker.io/...` vs `localhost/...` tag) → `retag`.
4. Localized “Contents” artifact leaking (e.g. French “Contenu”) → language-agnostic pre-heading strip; cleaned 70 pre-existing files.
5. Belarusian → Bulgarian (missing `languageName` entries) → engine fix + retests.

Pushed

- milosvasic.ru → efcef59 (canonical GitHub Pages repo).
- vasic.digital → 683a717.
- vasic (top) → c9a5daf (tooling, evidence, submodule bumps).
- HelixTranslate → 6ad2e8b (github Translator + HelixDevelopment).
- HelixConstitution → §11.4.140/§11.4.141 (pushed earlier, all 8 remotes).